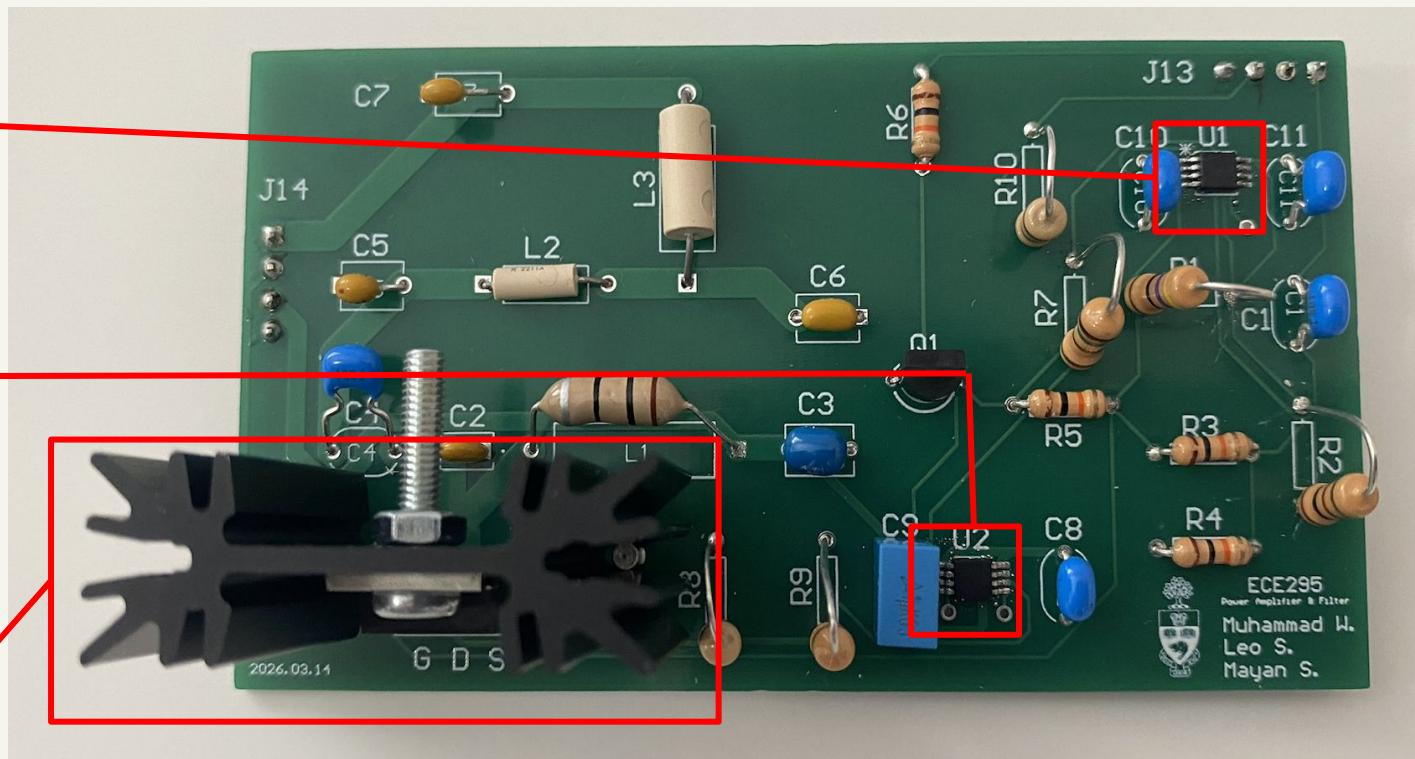


Class D Power Amplifier: Three Components

LTC6752
Comparator

MCP14A0602
Gate Driver

IRF510 Mosfet
(Amplifier)

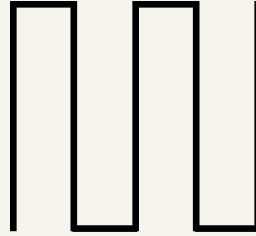
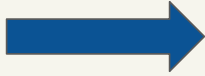


Note: **/TXEN** disables the drive path, shutting down the amplifier

LTC6752 Comparator: Converts the Sine Wave Into a Square Wave



Input Sine Wave



Output Square Wave

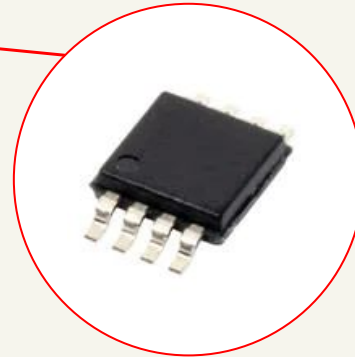
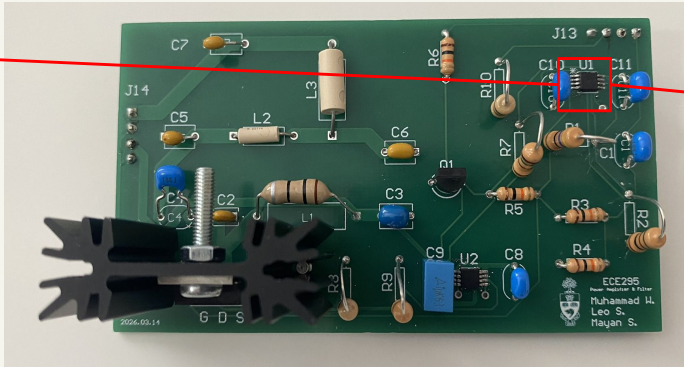
Role

Turns sinusoid into 0-5 V square wave

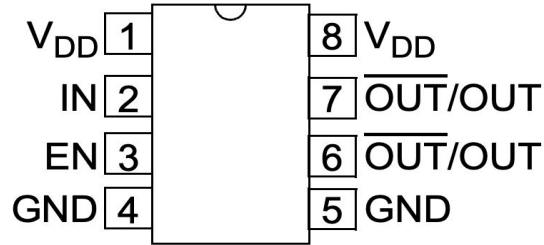
Why this part

Converts the analog input into the digital signal the gate driver needs [2]

LTC6752
Comparator



MCP14A0602 Gate Driver: Delivers High Current to Switch the MOSFET



Pin 3 is Connected to TXEN Control

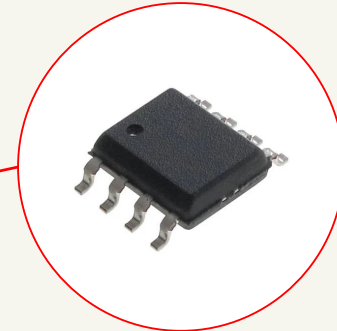
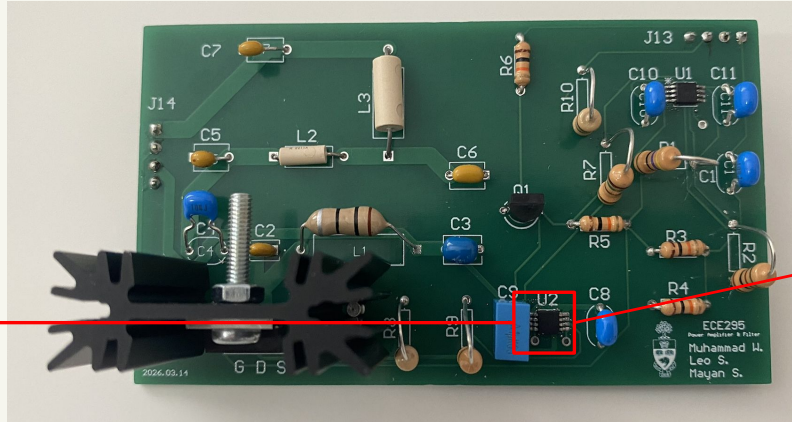
Requirement

The IRF510 gate requires 1.5-3 A of peak current to switch within 2-5 ns

Capability

Provides up to 6 A of peak current [3]

**MCP14A0602
Gate Driver**



IRF510 MOSFET: Produces the 10 W Amplifier Output

Current

5.6 A drain current provides a safe margin for 10 W

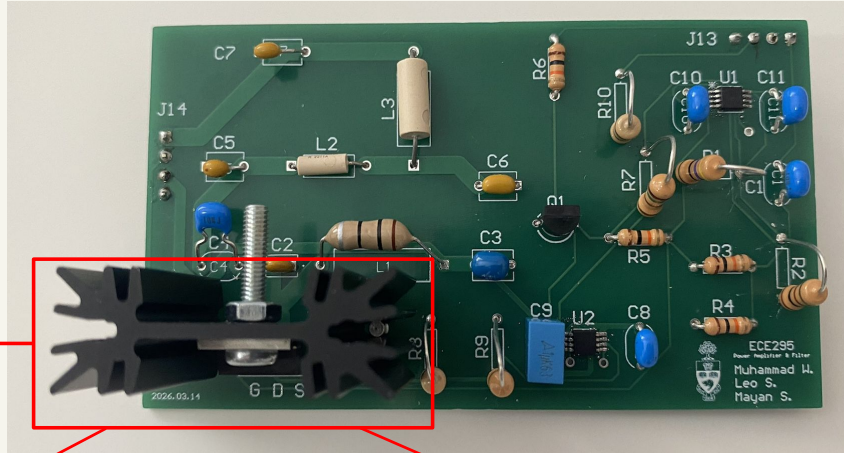
Voltage

100 V drain-source rating provides a safe margin above the 24 V drain swing

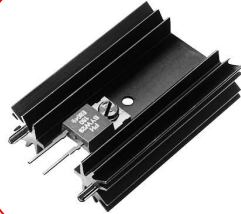
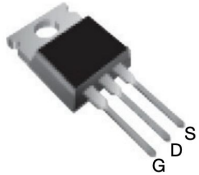
Package

TO-220 supports a heatsink for thermal management [4]

IRF510
Mosfet

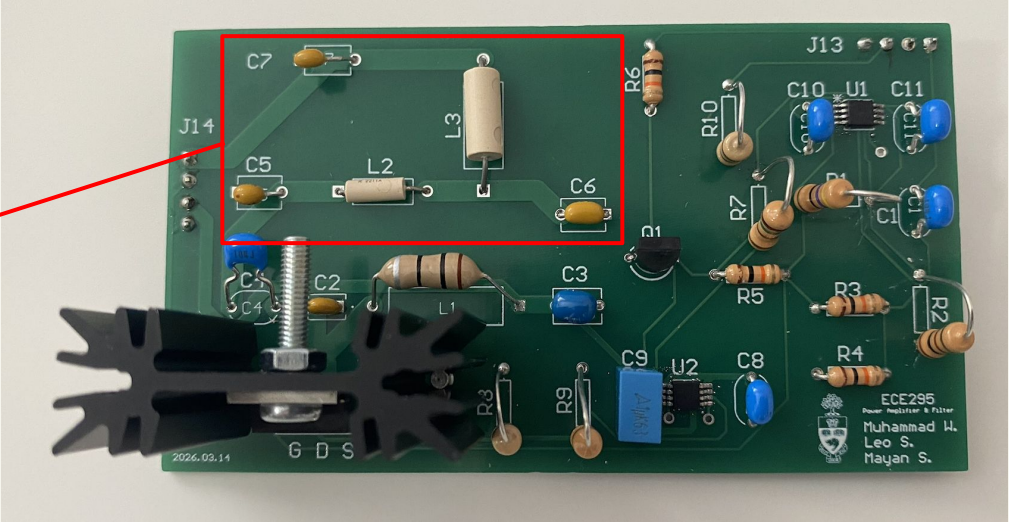
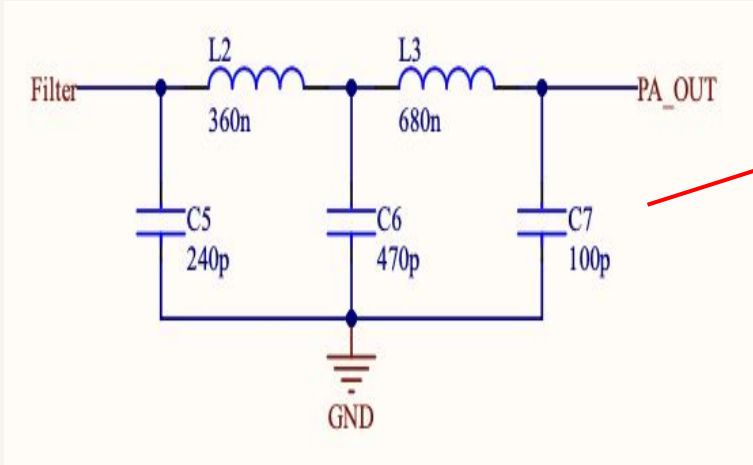


TO-220AB



Low Pass Filter: Removes Harmonics from the Amplifier Output

5th-order Butterworth (C-L-C-L-C)



Requirement	Target	Simulated Result	Outcome
THD (14 MHz)	< 10%	1.23%	MEETS